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Section: 5-B

Reg no: 2023-BSE-042

Lab 6

TASK NO 01:

Switch to root with su - and back to a normal user

STEP NO 01:

Set a root password (Ubuntu root is disabled by default; this enables su - temporarily for the lab):

```
manahil@ubuntu:~$ sudo passwd root
[sudo] password for manahil:
Sorry, try again.
[sudo] password for manahil:
New password:
Retype new password:
passwd: password updated successfully
manahil@ubuntu:~$ _
```

STEP NO 02:

Switch to root and verify:

```
manahil@ubuntu:~$ su -
Password:
root@ubuntu:~# whoami

root
root@ubuntu:~#
root@ubuntu:~# id
uid=0(root) gid=0(root) groups=0(root)
root@ubuntu:~# _
```

STEP NO 03:

Switch back to your normal user:

```
root@ubuntu:~# exit
logout
manahil@ubuntu:~$ whoami
manahil
manahil@ubuntu:~$ _
```

TASK NO 02:

Create user tom and verify in passwd/group/shadow

STEP NO 01:

Create user tom (interactive, sets password and home directory):

```
manahil
manahil@ubuntu:~$ sudo adduser tom


info: Adding user `tom' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `tom' (1002) ...
info: Adding new user `tom' (1002) with group `tom (1002)' ...
info: Creating home directory `/home/tom' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for tom
Enter the new value, or press ENTER for the default
        Full Name []: tom
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
info: Adding new user `tom' to supplemental / extra groups `users' ...
info: Adding user `tom' to group `users' ...
manahil@ubuntu:~$ _
```

STEP NO 02:

Verify tom in system files (view and visually confirm presence):

```
saned:x:115:120::/var/lib/saned:/usr/sbin/nologin
colord:x:116:121:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
pulse:x:117:122:PulseAudio daemon,,,:/run/pulse:/usr/sbin/nologin
cups-browsed:x:118:111::/nonexistent:/usr/sbin/nologin
xrdp:x:119:124::/run/xrdp:/usr/sbin/nologin
tom:x:1002:1002:tom,,,:/home/tom:/bin/bash
manahil@ubuntu:~$
```

```
pulse-access:x:123:
xrdp:x:124:
docker:x:988:manahil
tom:x:1002:
manahil@ubuntu:~$
manahil@ubuntu:~$
```

```
cups-browsed:!:20385::::::
xrdp:!:20385::::::
tom:$y$j9T$jdI5mSI2.PAc2hGMXHOUO.$8JVk.1k.pcYsYLAkFhl9THP28u/pA.rLC3oAbW57oaB:20391:0:99999:7:::
manahil@ubuntu:~$ _
```

TASK NO 03:

Create groups; change tom's primary and secondary groups

STEP NO 01:

Create groups and verify by viewing /etc/group (visually confirm entries exist):

```
manahil@ubuntu:~$ sudo groupadd developer
manahil@ubuntu:~$ sudo groupadd devops
manahil@ubuntu:~$ sudo groupadd designer
manahil@ubuntu:~$
```

```
manahil@ubuntu:~$ sudo usermod -aG developer,devops tom

manahil@ubuntu:~$ id tom
uid=1002(tom) gid=1005(designer) groups=1005(designer),100(users),1003(developer),1004(devops)
manahil@ubuntu:~$ groups tom
tom : designer users developer devops
manahil@ubuntu:~$
```

STEP NO 02:

Change tom's primary group to designer and verify:

```
manahil@ubuntu:~$ sudo usermod -g designer tom

manahil@ubuntu:~$ id tom
uid=1002(tom) gid=1005(designer) groups=1005(designer),100(users)
manahil@ubuntu:~$
manahil@ubuntu:~$ _
```

STEP NO 03:

Add secondary groups developer and devops to tom and verify:

```
manahil@ubuntu:~$ groups tom
tom : designer users developer devops
manahil@ubuntu:~$ _
```

```
manahil@ubuntu:~$ sudo usermod -aG developer,devops tom

manahil@ubuntu:~$ id tom
uid=1002(tom) gid=1005(designer) groups=1005(designer),100(users),1003(developer),1004(devops)
manahil@ubuntu:~$ groups tom
tom : designer users developer devops
manahil@ubuntu:~$
```

STEP NO 04:

Replace all secondary groups so only tom (user's own group) remains and verify:

```
manahil@ubuntu:~$ sudo usermod -G tom tom

manahil@ubuntu:~$ id tom
uid=1002(tom) gid=1005(designer) groups=1005(designer),1002(tom)
manahil@ubuntu:~$ groups tom
tom : designer tom
manahil@ubuntu:~$ _
```

TASK NO 04:

Create/delete users (Jerry, Scooby) and groups (jolly, anime)

STEP NO 01:

Create users:

```
manahil@ubuntu:~$ sudo adduser jerry
info: Adding user `jerry' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `jerry' (1007) ...
info: Adding new user `jerry' (1007) with group `jerry (1007)' ...
info: Creating home directory `/home/jerry' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for jerry
Enter the new value, or press ENTER for the default
        Full Name []: jerry
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
info: Adding new user `jerry' to supplemental / extra groups `users' ...
info: Adding user `jerry' to group `users' ...
manahil@ubuntu:~$ sudo useradd scooby
manahil@ubuntu:~$
```

STEP NO 02:

Try to log in as Scooby immediately (expected authentication failure because there is no password yet):

```
manahil@ubuntu:~$ su - Scooby
Password:
su: Authentication failure
manahil@ubuntu:~$ _
```

STEP NO 03:

Set a password for Scooby:

```
manahil@ubuntu:~$ sudo passwd Scooby

New password:
Retype new password:
passwd: password updated successfully
manahil@ubuntu:~$
```

STEP NO 04:

Try logging in as Scooby again (home directory still missing; expect a message such as“No directory, logging in with HOME=/”):

```
manahil@ubuntu:~$ su - Scooby
Password:
su: warning: cannot change directory to /home/Scooby: No such file or directory
$
```

STEP NO 05:

Show that Scooby’s home directory does not exist yet and what /etc/passwd says:

```
tom:x:1002:1005:tom,,,:/home/tom:/bin/bash
Scooby:x:1003:1006::/home/Scooby:/bin/sh
jerry:x:1007:1007:jerry,,,:/home/jerry:/bin/bash
scooby:x:1008:1008::/home/scooby:/bin/sh
manahil@ubuntu:~$ ls -ld /home/Scooby
ls: cannot access '/home/Scooby': No such file or directory
manahil@ubuntu:~$
```

STEP NO 06:

Manually create Scooby’s home directory and set proper ownership and permissions:

```
manahil@ubuntu:~$ sudo mkdir -p /home/Scooby
manahil@ubuntu:~$ sudo chown Scooby:Scooby /home/Scooby
manahil@ubuntu:~$ sudo chmod 750 /home/Scooby
manahil@ubuntu:~$ ls -ld /home/Scooby

drwxr-x--- 2 Scooby Scooby 4096 Oct 30 10:35 /home/Scooby
manahil@ubuntu:~$
manahil@ubuntu:~$
```

STEP NO 07:

Log in as Scooby again and verify you land in the correct home directory:

```
manahil@ubuntu:~$ su - Scooby
Password:
$ pwd
/home/Scooby
$ ls -la
total 8
drwxr-x--- 2 Scooby Scooby 4096 Oct 30 10:35 .
drwxr-xr-x 7 root   root   4096 Oct 30 10:35 ..
$ _
```

STEP NO 08:

Verify users in system files:

```
xrdp:x:119:124::/run/xrdp:/usr/sbin/nologin
tom:x:1002:1005:tom,,,:/home/tom:/bin/bash
Scooby:x:1003:1006::/home/Scooby:/bin/sh
jerry:x:1007:1007:jerry,,,:/home/jerry:/bin/bash
scooby:x:1008:1008::/home/scooby:/bin/sh
manahil@ubuntu:~$
manahil@ubuntu:~$ _
```

STEP NO 09:

Create groups:

```
manahil@ubuntu:~$ sudo addgroup jolly
info: Selecting GID from range 1000 to 59999 ...
info: Adding group `jolly' (GID 1009) ...
manahil@ubuntu:~$ sudo groupadd anime
manahil@ubuntu:~$
```

STEP NO 10:

Verify groups:

```
jerry:x:1007:
scooby:x:1008:
jolly:x:1009:
anime:x:1010:
manahil@ubuntu:~$
manahil@ubuntu:~$ _
```

STEP NO 11:

Delete groups and users:

```
manahil@ubuntu:~$ sudo deluser --remove-home jerry
info: Looking for files to backup/remove ...
info: Removing files ...
info: Removing crontab ...
info: Removing user `jerry' ...
manahil@ubuntu:~$ sudo userdel -r scooby
```

```
whoopsie:x:113:117::/nonexistent:/bin/false
avahi:x:114:118:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
saned:x:115:120::/var/lib/saned:/usr/sbin/nologin
colord:x:116:121:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
pulse:x:117:122:PulseAudio daemon,,,:/run/pulse:/usr/sbin/nologin
cups-browsed:x:118:111::/nonexistent:/usr/sbin/nologin
xrdp:x:119:124::/run/xrdp:/usr/sbin/nologin
tom:x:1002:1005:tom,,,:/home/tom:/bin/bash
manahil@ubuntu:~$
```

TASK NO 05:

Create user Student; create files; set owner/group; identify file types

STEP NO 01:

Create Student:

```
            NAME_REGEX in configuration.
manahil@ubuntu:~$ sudo adduser student

info: Adding user `student' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `student' (1006) ...
info: Adding new user `student' (1006) with group `student (1006)' ...
info: Creating home directory `/home/student' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for student
Enter the new value, or press ENTER for the default
        Full Name []: student
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
info: Adding new user `student' to supplemental / extra groups `users' ...
info: Adding user `student' to group `users' ...
manahil@ubuntu:~$ _
```

STEP NO 02:

Switch to Student and create files:

```
manahil@ubuntu:~$ su - student
Password:
student@ubuntu:~$ touch file1
student@ubuntu:~$ mkdir -p dir1
student@ubuntu:~$ touch dir1/file2
student@ubuntu:~$ ls -l
total 4
drwxrwxr-x 2 student student 4096 Oct 30 14:57 dir1
-rw-rw-r-- 1 student student    0 Oct 30 14:57 file1
student@ubuntu:~$
```

STEP NO 03:

Change owner then group for file1 (separate commands):

```
manahil@ubuntu:~$ sudo chown tom /home/student/file1
manahil@ubuntu:~$ ls -l /home/student/file1
```

```
student@ubuntu:~$ ls -l /home/student/file1
-rw-rw-r-- 1 tom student 0 Oct 31 13:06 /home/student/file1
student@ubuntu:~$ _
```

```
student@ubuntu:/home/manahil$ sudo chgrp devops /home/Student/file1

[sudo] password for student:
313Sorry, try again.
[sudo] password for student:
student is not in the sudoers file.
student@ubuntu:/home/manahil$ ls -l /home/student/file1
-rw-rw-r-- 1 tom devops 0 Oct 31 13:06 /home/student/file1
student@ubuntu:/home/manahil$ _
```

STEP NO 04:

Identify files/directories and show /dev/null:

```
manahil@ubuntu:~$ sudo ls -l /home/student/file1
-rw-rw-r-- 1 tom devops 0 Oct 31 13:06 /home/student/file1
manahil@ubuntu:~$ sudo ls -l /home/student/dir1
total 0
-rw-rw-r-- 1 student student 0 Oct 31 13:06 file2
manahil@ubuntu:~$ sudo ls -l /dev/null
crw-rw-rw- 1 root root 1, 3 Oct 30 09:05 /dev/null
manahil@ubuntu:~$ sudo file /home/student/file1 /home/student/dir1 /dev/null
/home/student/file1: empty
/home/student/dir1:  directory
/dev/null:           character special (1/3)
manahil@ubuntu:~$ _
```

TASK NO 06:

Change permissions using symbolic mode

```
manahil@ubuntu:~$ su - student
Password:
student@ubuntu:~$ cd ~
student@ubuntu:~$ ls -l file1
-rw-rw-r-- 1 tom devops 0 Oct 31 13:06 file1
student@ubuntu:~$ _
```

STEP NO 01:

Ensure Student and file present:

```
manahil@ubuntu:~$ sudo chown student:student /home/student/file1
[sudo] password for manahil:
Sorry, try again.
[sudo] password for manahil:
Sorry, try again.
[sudo] password for manahil:
manahil@ubuntu:~$ su - student
Password:
student@ubuntu:~$ ls -l file1
-rw-rw-r-- 1 student student 0 Oct 31 13:06 file1
student@ubuntu:~$
```

STEP NO 02:

Remove all permissions:

```
student@ubuntu:~$ chmod -rwx file1
student@ubuntu:~$ ls -l file1

---------- 1 student student 0 Oct 31 13:06 file1
student@ubuntu:~$
student@ubuntu:~$ _
```

STEP NO 03:

Add read to all:

```
student@ubuntu:~$ chmod +r file1
student@ubuntu:~$ ls -l file1
-r--r--r-- 1 student student 0 Oct 31 13:06 file1
student@ubuntu:~$
```

STEP NO 04:

Add execute to user:

```
student@ubuntu:~$ chmod u+x file1
student@ubuntu:~$ ls -l file1
-r-xr--r-- 1 student student 0 Oct 31 13:06 file1
student@ubuntu:~$
```

STEP NO 05:

Add write to user and group:

```
student@ubuntu:~$ chmod ug+w file1
student@ubuntu:~$ ls -l file1
-rwxrw-r-- 1 student student 0 Oct 31 13:06 file1
student@ubuntu:~$ _
```

STEP NO 06:

Remove all permissions (explicit):

```
student@ubuntu:~$ chmod ugo-rwx file1
student@ubuntu:~$ ls -l file1
---------- 1 student student 0 Oct 31 13:06 file1
student@ubuntu:~$ _
```

TASK NO 07:

Change permissions using “set” symbolic form (u= g= o=)

Ensure you are Student:

```
student@ubuntu:~$ su - student
Password:
student@ubuntu:~$ cd ~
student@ubuntu:~$ ls -l file1
---------- 1 student student 0 Oct 31 13:06 file1
student@ubuntu:~$ _
```

STEP NO 01:

Set all to rwx:

```
---------- 1 student student 0 Oct 31 13:06 file1
student@ubuntu:~$ chmod u=rwx,g=rwx,o=rwx file1
student@ubuntu:~$ ls -l file1
-rwxrwxrwx 1 student student 0 Oct 31 13:06 file1
student@ubuntu:~$
```

STEP NO 02:

Remove execute from group and others:

```
student@ubuntu:~$ chmod g=rw,o=rw file1
student@ubuntu:~$ ls -l file1
-rwxrw-rw- 1 student student 0 Oct 31 13:06 file1
student@ubuntu:~$ _
```

STEP NO 03:

Remove all permissions:

```
student@ubuntu:~$ chmod u=,g=,o= file1
student@ubuntu:~$ ls -l file1
---------- 1 student student 0 Oct 31 13:06 file1
student@ubuntu:~$
```

TASK NO 08:

Change permissions using numeric (octal) mode

Ensure you are Student:

```
student@ubuntu:~$ su - student
Password:
student@ubuntu:~$ cd ~
student@ubuntu:~$ ls -l file1
---------- 1 student student 0 Oct 31 13:06 file1
student@ubuntu:~$
```

Run each command and capture screenshot after each ls:

STEP NO 01:

```
student@ubuntu:~$ chmod 777 file1
student@ubuntu:~$
student@ubuntu:~$ ls -l file1
-rwxrwxrwx 1 student student 0 Oct 31 13:06 file1
student@ubuntu:~$ _
```

STEP NO 02:

```
student@ubuntu:~$ chmod 700 file1

student@ubuntu:~$
student@ubuntu:~$ ls -l file1
-rwx------ 1 student student 0 Oct 31 13:06 file1
student@ubuntu:~$ _
```

STEP NO 03:

```
student@ubuntu:~$ chmod 744 file1
student@ubuntu:~$ ls -l file1
-rwxr--r-- 1 student student 0 Oct 31 13:06 file1
student@ubuntu:~$ _
```

STEP NO 04:

```
student@ubuntu:~$ chmod 640 file1
student@ubuntu:~$ ls -l file1
-rw-r----- 1 student student 0 Oct 31 13:06 file1
student@ubuntu:~$ _
```

STEP NO 05

```
student@ubuntu:~$ chmod 664 file1
student@ubuntu:~$ ls -l file1
-rw-rw-r-- 1 student student 0 Oct 31 13:06 file1
student@ubuntu:~$
```

STEP NO 06:

```
student@ubuntu:~$ chmod 775 file1
student@ubuntu:~$ ls -l file1
-rwxrwxr-x 1 student student 0 Oct 31 13:06 file1
student@ubuntu:~$
```

STEP NO 07:

```
student@ubuntu:~$ chmod 750 file1
student@ubuntu:~$ ls -l file1
-rwxr-x--- 1 student student 0 Oct 31 13:06 file1
student@ubuntu:~$
```

TASK NO 09:

Practice pipes, pagers, grep, and redirects with /var/log/syslog

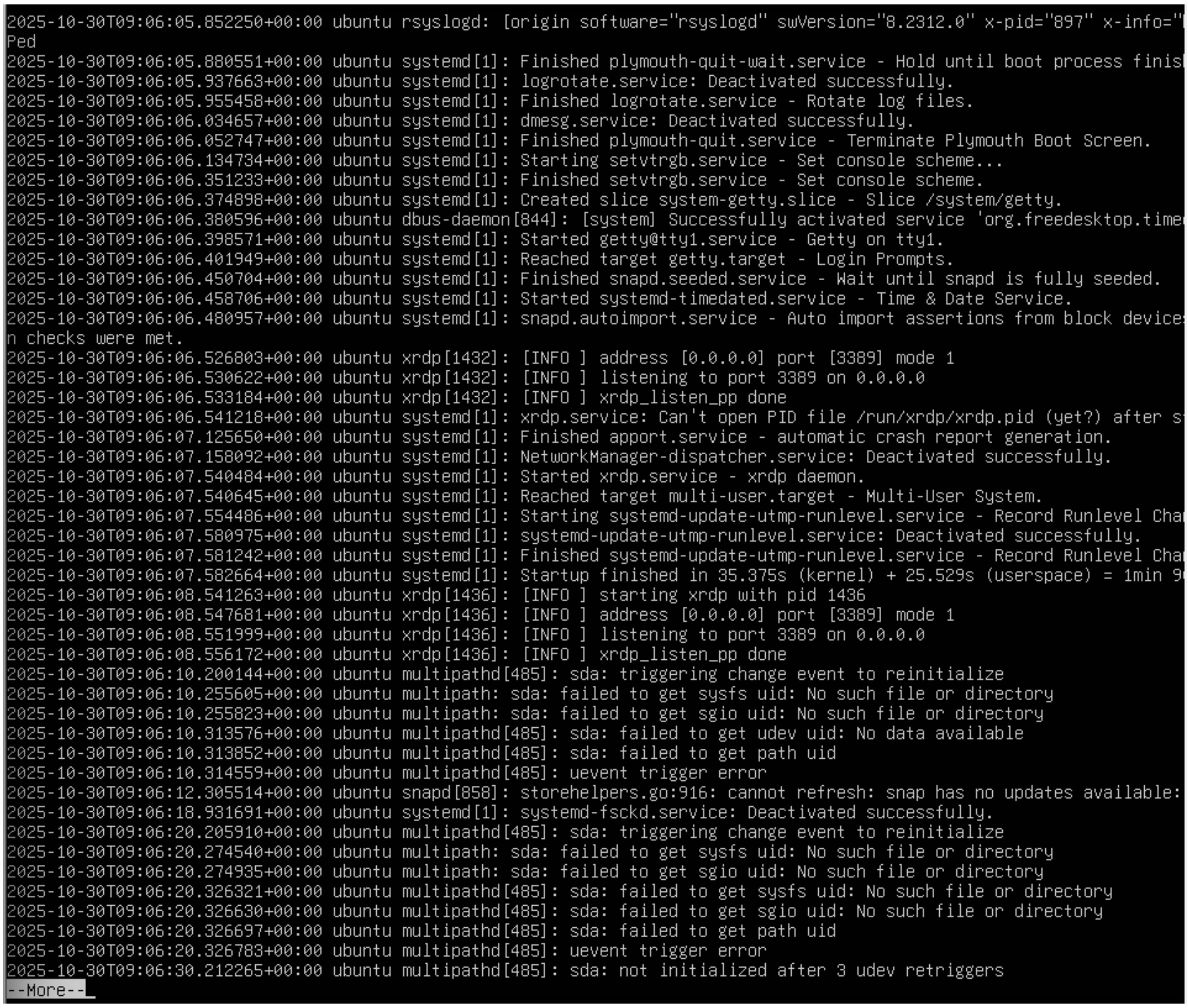
STEP NO 01:

less:

```
2025-10-30T09:06:05.848889+00:00 ubuntu systemd[1]: rsyslog.service: Sent signal SIGHUP to main process 897 (rsyslo
2025-10-30T09:06:05.852250+00:00 ubuntu rsyslogd: [origin software="rsyslogd" swVersion="8.2312.0" x-pid="897" x-in
Ped
2025-10-30T09:06:05.880551+00:00 ubuntu systemd[1]: Finished plymouth-quit-wait.service - Hold until boot process f
2025-10-30T09:06:05.937663+00:00 ubuntu systemd[1]: logrotate.service: Deactivated successfully.
2025-10-30T09:06:05.955458+00:00 ubuntu systemd[1]: Finished logrotate.service - Rotate log files.
2025-10-30T09:06:06.034657+00:00 ubuntu systemd[1]: dmesg.service: Deactivated successfully.
2025-10-30T09:06:06.052747+00:00 ubuntu systemd[1]: Finished plymouth-quit.service - Terminate Plymouth Boot Screen
2025-10-30T09:06:06.134734+00:00 ubuntu systemd[1]: Starting setvtrgb.service - Set console scheme...
2025-10-30T09:06:06.351233+00:00 ubuntu systemd[1]: Finished setvtrgb.service - Set console scheme.
2025-10-30T09:06:06.374898+00:00 ubuntu systemd[1]: Created slice system-getty.slice - Slice /system/getty.
2025-10-30T09:06:06.380596+00:00 ubuntu dbus-daemon[844]: [system] Successfully activated service 'org.freedesktop.
2025-10-30T09:06:06.398571+00:00 ubuntu systemd[1]: Started getty@tty1.service - Getty on tty1.
2025-10-30T09:06:06.401949+00:00 ubuntu systemd[1]: Reached target getty.target - Login Prompts.
2025-10-30T09:06:06.450704+00:00 ubuntu systemd[1]: Finished snapd.seeded.service - Wait until snapd is fully seede
2025-10-30T09:06:06.458706+00:00 ubuntu systemd[1]: Started systemd-timedated.service - Time & Date Service.
2025-10-30T09:06:06.480957+00:00 ubuntu systemd[1]: snapd.autoimport.service - Auto import assertions from block de
n checks were met.
2025-10-30T09:06:06.526803+00:00 ubuntu xrdp[1432]: [INFO ] address [0.0.0.0] port [3389] mode 1
2025-10-30T09:06:06.530622+00:00 ubuntu xrdp[1432]: [INFO ] listening to port 3389 on 0.0.0.0
2025-10-30T09:06:06.533184+00:00 ubuntu xrdp[1432]: [INFO ] xrdp_listen_pp done
2025-10-30T09:06:06.541218+00:00 ubuntu systemd[1]: xrdp.service: Can't open PID file /run/xrdp/xrdp.pid (yet?) aft
2025-10-30T09:06:07.125650+00:00 ubuntu systemd[1]: Finished apport.service - automatic crash report generation.
2025-10-30T09:06:07.158092+00:00 ubuntu systemd[1]: NetworkManager-dispatcher.service: Deactivated successfully.
2025-10-30T09:06:07.540484+00:00 ubuntu systemd[1]: Started xrdp.service - xrdp daemon.
2025-10-30T09:06:07.540645+00:00 ubuntu systemd[1]: Reached target multi-user.target - Multi-User System.
2025-10-30T09:06:07.554486+00:00 ubuntu systemd[1]: Starting systemd-update-utmp-runlevel.service - Record Runlevel
2025-10-30T09:06:07.580975+00:00 ubuntu systemd[1]: systemd-update-utmp-runlevel.service: Deactivated successfully.
2025-10-30T09:06:07.581242+00:00 ubuntu systemd[1]: Finished systemd-update-utmp-runlevel.service - Record Runlevel
2025-10-30T09:06:07.582664+00:00 ubuntu systemd[1]: Startup finished in 35.375s (kernel) + 25.529s (userspace) = 1m
2025-10-30T09:06:08.541263+00:00 ubuntu xrdp[1436]: [INFO ] starting xrdp with pid 1436
2025-10-30T09:06:08.547681+00:00 ubuntu xrdp[1436]: [INFO ] address [0.0.0.0] port [3389] mode 1
2025-10-30T09:06:08.551999+00:00 ubuntu xrdp[1436]: [INFO ] listening to port 3389 on 0.0.0.0
2025-10-30T09:06:08.556172+00:00 ubuntu xrdp[1436]: [INFO ] xrdp_listen_pp done
2025-10-30T09:06:10.200144+00:00 ubuntu multipathd[485]: sda: triggering change event to reinitialize
2025-10-30T09:06:10.255605+00:00 ubuntu multipath: sda: failed to get sysfs uid: No such file or directory
2025-10-30T09:06:10.255823+00:00 ubuntu multipath: sda: failed to get sgio uid: No such file or directory
2025-10-30T09:06:10.313576+00:00 ubuntu multipathd[485]: sda: failed to get udev uid: No data available
2025-10-30T09:06:10.313852+00:00 ubuntu multipathd[485]: sda: failed to get path uid
2025-10-30T09:06:10.314559+00:00 ubuntu multipathd[485]: uevent trigger error
2025-10-30T09:06:12.305514+00:00 ubuntu snapd[858]: storehelpers.go:916: cannot refresh: snap has no updates availa
2025-10-30T09:06:18.931691+00:00 ubuntu systemd[1]: systemd-fsckd.service: Deactivated successfully.
2025-10-30T09:06:20.205910+00:00 ubuntu multipathd[485]: sda: triggering change event to reinitialize
2025-10-30T09:06:20.274540+00:00 ubuntu multipath: sda: failed to get sysfs uid: No such file or directory
2025-10-30T09:06:20.274935+00:00 ubuntu multipath: sda: failed to get sgio uid: No such file or directory
2025-10-30T09:06:20.326321+00:00 ubuntu multipathd[485]: sda: failed to get sysfs uid: No such file or directory
2025-10-30T09:06:20.326630+00:00 ubuntu multipathd[485]: sda: failed to get sgio uid: No such file or directory
2025-10-30T09:06:20.326697+00:00 ubuntu multipathd[485]: sda: failed to get path uid
2025-10-30T09:06:20.326783+00:00 ubuntu multipathd[485]: uevent trigger error
:_
```

STEP NO 02:

more:



```
2025-10-30T09:06:05.852250+00:00 ubuntu rsyslogd: [origin software="rsyslogd" swVersion="8.2312.0" x-pid="897" x-info="
Ped
2025-10-30T09:06:05.880551+00:00 ubuntu systemd[1]: Finished plymouth-quit-wait.service - Hold until boot process finis
2025-10-30T09:06:05.937663+00:00 ubuntu systemd[1]: logrotate.service: Deactivated successfully.
2025-10-30T09:06:05.955458+00:00 ubuntu systemd[1]: Finished logrotate.service - Rotate log files.
2025-10-30T09:06:06.034657+00:00 ubuntu systemd[1]: dmesg.service: Deactivated successfully.
2025-10-30T09:06:06.052747+00:00 ubuntu systemd[1]: Finished plymouth-quit.service - Terminate Plymouth Boot Screen.
2025-10-30T09:06:06.134734+00:00 ubuntu systemd[1]: Starting setvtrgb.service - Set console scheme...
2025-10-30T09:06:06.351233+00:00 ubuntu systemd[1]: Finished setvtrgb.service - Set console scheme.
2025-10-30T09:06:06.374898+00:00 ubuntu systemd[1]: Created slice system-getty.slice - Slice /system/getty.
2025-10-30T09:06:06.380596+00:00 ubuntu dbus-daemon[844]: [system] Successfully activated service 'org.freedesktop.time
2025-10-30T09:06:06.398571+00:00 ubuntu systemd[1]: Started getty@tty1.service - Getty on tty1.
2025-10-30T09:06:06.401949+00:00 ubuntu systemd[1]: Reached target getty.target - Login Prompts.
2025-10-30T09:06:06.450704+00:00 ubuntu systemd[1]: Finished snapd.seeded.service - Wait until snapd is fully seeded.
2025-10-30T09:06:06.458706+00:00 ubuntu systemd[1]: Started systemd-timedated.service - Time & Date Service.
2025-10-30T09:06:06.480957+00:00 ubuntu systemd[1]: snapd.autoimport.service - Auto import assertions from block device
n checks were met.
2025-10-30T09:06:06.526803+00:00 ubuntu xrdp[1432]: [INFO ] address [0.0.0.0] port [3389] mode 1
2025-10-30T09:06:06.530622+00:00 ubuntu xrdp[1432]: [INFO ] listening to port 3389 on 0.0.0.0
2025-10-30T09:06:06.533184+00:00 ubuntu xrdp[1432]: [INFO ] xrdp_listen_pp done
2025-10-30T09:06:06.541218+00:00 ubuntu systemd[1]: xrdp.service: Can't open PID file /run/xrdp/xrdp.pid (yet?) after s
2025-10-30T09:06:07.125650+00:00 ubuntu systemd[1]: Finished apport.service - automatic crash report generation.
2025-10-30T09:06:07.158092+00:00 ubuntu systemd[1]: NetworkManager-dispatcher.service: Deactivated successfully.
2025-10-30T09:06:07.540484+00:00 ubuntu systemd[1]: Started xrdp.service - xrdp daemon.
2025-10-30T09:06:07.540645+00:00 ubuntu systemd[1]: Reached target multi-user.target - Multi-User System.
2025-10-30T09:06:07.554486+00:00 ubuntu systemd[1]: Starting systemd-update-utmp-runlevel.service - Record Runlevel Cha
2025-10-30T09:06:07.580975+00:00 ubuntu systemd[1]: systemd-update-utmp-runlevel.service: Deactivated successfully.
2025-10-30T09:06:07.581242+00:00 ubuntu systemd[1]: Finished systemd-update-utmp-runlevel.service - Record Runlevel Cha
2025-10-30T09:06:07.582664+00:00 ubuntu systemd[1]: Startup finished in 35.375s (kernel) + 25.529s (userspace) = 1min 9
2025-10-30T09:06:08.541263+00:00 ubuntu xrdp[1436]: [INFO ] starting xrdp with pid 1436
2025-10-30T09:06:08.547681+00:00 ubuntu xrdp[1436]: [INFO ] address [0.0.0.0] port [3389] mode 1
2025-10-30T09:06:08.551999+00:00 ubuntu xrdp[1436]: [INFO ] listening to port 3389 on 0.0.0.0
2025-10-30T09:06:08.556172+00:00 ubuntu xrdp[1436]: [INFO ] xrdp_listen_pp done
2025-10-30T09:06:10.200144+00:00 ubuntu multipathd[485]: sda: triggering change event to reinitialize
2025-10-30T09:06:10.255605+00:00 ubuntu multipath: sda: failed to get sysfs uid: No such file or directory
2025-10-30T09:06:10.255823+00:00 ubuntu multipath: sda: failed to get sgio uid: No such file or directory
2025-10-30T09:06:10.313576+00:00 ubuntu multipathd[485]: sda: failed to get udev uid: No data available
2025-10-30T09:06:10.313852+00:00 ubuntu multipathd[485]: sda: failed to get path uid
2025-10-30T09:06:10.314559+00:00 ubuntu multipathd[485]: uevent trigger error
2025-10-30T09:06:12.305514+00:00 ubuntu snapd[858]: storehelpers.go:916: cannot refresh: snap has no updates available:
2025-10-30T09:06:18.931691+00:00 ubuntu systemd[1]: systemd-fsckd.service: Deactivated successfully.
2025-10-30T09:06:20.205910+00:00 ubuntu multipathd[485]: sda: triggering change event to reinitialize
2025-10-30T09:06:20.274540+00:00 ubuntu multipath: sda: failed to get sysfs uid: No such file or directory
2025-10-30T09:06:20.274935+00:00 ubuntu multipath: sda: failed to get sgio uid: No such file or directory
2025-10-30T09:06:20.326321+00:00 ubuntu multipathd[485]: sda: failed to get sysfs uid: No such file or directory
2025-10-30T09:06:20.326630+00:00 ubuntu multipathd[485]: sda: failed to get sgio uid: No such file or directory
2025-10-30T09:06:20.326697+00:00 ubuntu multipathd[485]: sda: failed to get path uid
2025-10-30T09:06:20.326783+00:00 ubuntu multipathd[485]: uevent trigger error
2025-10-30T09:06:30.212265+00:00 ubuntu multipathd[485]: sda: not initialized after 3 udev retriggers
--More--
```

STEP NO 03:

grep failures/errors:

```
2025-10-30T09:21:26.860822+00:00 ubuntu avahi-daemon[843]: New relevant interface ens33.IPv4 for mDNS.
2025-10-30T09:21:26.862992+00:00 ubuntu systemd-timesyncd[681]: Network configuration changed, trying to establish conn
2025-10-30T09:21:26.863132+00:00 ubuntu avahi-daemon[843]: Registering new address record for 192.168.58.139 on ens33.I
2025-10-30T09:21:27.069987+00:00 ubuntu systemd-timesyncd[681]: Contacted time server 185.125.190.58:123 (ntp.ubuntu.co
2025-10-30T09:23:02.428093+00:00 ubuntu systemd[1600]: launchpadlib-cache-clean.service - Clean up old files in the Lau
 unmet condition check (ConditionPathExists=/home/manahil/.launchpadlib/api.launchpad.net/cache).
2025-10-30T09:25:01.766264+00:00 ubuntu CRON[2969]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
2025-10-30T09:26:21.281691+00:00 ubuntu kernel: workqueue: blk_mq_run_work_fn hogged CPU for >13333us 16 times, conside
2025-10-30T09:26:25.423558+00:00 ubuntu kernel: workqueue: hub_event hogged CPU for >13333us 4 times, consider switchin
2025-10-30T09:28:45.165234+00:00 ubuntu kernel: workqueue: e1000_watchdog [e1000] hogged CPU for >13333us 4 times, cons
2025-10-30T09:30:02.452799+00:00 ubuntu systemd[1]: Starting sysstat-collect.service - system activity accounting tool.
2025-10-30T09:30:02.554152+00:00 ubuntu systemd[1]: sysstat-collect.service: Deactivated successfully.
2025-10-30T09:30:02.555686+00:00 ubuntu systemd[1]: Finished sysstat-collect.service - system activity accounting tool.
2025-10-30T09:30:56.683081+00:00 ubuntu PackageKit: daemon quit
2025-10-30T09:30:56.704325+00:00 ubuntu systemd[1]: packagekit.service: Deactivated successfully.
2025-10-30T09:33:04.446676+00:00 ubuntu systemd[1]: Starting apt-news.service - Update APT News...
2025-10-30T09:33:04.498144+00:00 ubuntu systemd[1]: Starting esm-cache.service - Update the local ESM caches...
2025-10-30T09:33:04.635446+00:00 ubuntu kernel: kauditd_printk_skb: 93 callbacks suppressed
2025-10-30T09:33:04.635491+00:00 ubuntu kernel: audit: type=1400 audit(1761816784.633:340): apparmor="DENIED" operation
_apt_news" name="/usr/lib/python3/dist-packages/uaclient/__pycache__/__init__.cpython-312.pyc.129326029928240" pid=3205
_mask="c" fsuid=0 ouid=0
2025-10-30T09:33:04.701424+00:00 ubuntu kernel: audit: type=1400 audit(1761816784.693:341): apparmor="DENIED" operation
_apt_news" name="/usr/lib/python3/dist-packages/uaclient/__pycache__/apt.cpython-312.pyc.129326030102000" pid=3205 comm
="c" fsuid=0 ouid=0
2025-10-30T09:33:04.742419+00:00 ubuntu kernel: audit: type=1400 audit(1761816784.741:342): apparmor="DENIED" operation
_esm_cache" name="/usr/lib/python3/dist-packages/uaclient/__pycache__/__init__.cpython-312.pyc.130056052952496" pid=320
d_mask="c" fsuid=0 ouid=0
2025-10-30T09:33:04.750420+00:00 ubuntu kernel: audit: type=1400 audit(1761816784.748:343): apparmor="DENIED" operation
_esm_cache" name="/usr/lib/python3/dist-packages/uaclient/__pycache__/log.cpython-312.pyc.130056057649536" pid=3206 com
k="c" fsuid=0 ouid=0
2025-10-30T09:33:04.780418+00:00 ubuntu kernel: audit: type=1400 audit(1761816784.779:344): apparmor="DENIED" operation
_esm_cache" name="/usr/lib/python3/dist-packages/uaclient/__pycache__/defaults.cpython-312.pyc.130056051841072" pid=320
d_mask="c" fsuid=0 ouid=0
2025-10-30T09:33:04.782408+00:00 ubuntu kernel: audit: type=1400 audit(1761816784.781:345): apparmor="DENIED" operation
_esm_cache" name="/usr/lib/python3/dist-packages/uaclient/__pycache__/secret_manager.cpython-312.pyc.130056051841072" p
 denied_mask="c" fsuid=0 ouid=0
2025-10-30T09:33:04.801429+00:00 ubuntu kernel: audit: type=1400 audit(1761816784.799:346): apparmor="DENIED" operation
_esm_cache" name="/usr/lib/python3/dist-packages/uaclient/__pycache__/system.cpython-312.pyc.130056051843120" pid=3206
mask="c" fsuid=0 ouid=0
2025-10-30T09:33:04.871417+00:00 ubuntu kernel: audit: type=1400 audit(1761816784.870:347): apparmor="DENIED" operation
_esm_cache" name="/usr/lib/python3/dist-packages/uaclient/__pycache__/exceptions.cpython-312.pyc.130056052272688" pid=3
ied_mask="c" fsuid=0 ouid=0
2025-10-30T09:33:04.875421+00:00 ubuntu kernel: audit: type=1400 audit(1761816784.874:348): apparmor="DENIED" operation
_apt_news" name="/usr/lib/python3/dist-packages/uaclient/__pycache__/event_logger.cpython-312.pyc.129326025135792" pid=
nied_mask="c" fsuid=0 ouid=0
2025-10-30T09:33:04.882411+00:00 ubuntu kernel: audit: type=1400 audit(1761816784.881:349): apparmor="DENIED" operation
/var/log/syslog | head
Pattern not found
manahil@ubuntu:~$
```

STEP NO 04:

redirect:

append:

```
manahil@ubuntu:~$ sudo grep -i systemd /var/log/syslog > ~/syslog_systemd.txt

grep: /var/log/syslog: binary file matches
manahil@ubuntu:~$ sudo grep -i network /var/log/syslog >> ~/syslog_systemd.txt
grep: /var/log/syslog: binary file matches
manahil@ubuntu:~$
```

TASK NO 10:

Script setup.sh – variables, command substitution, file/dir checks, permissions (use vim)

1. Include bash shebang

```
#!/bin/bash
#
#
```

```
~
"setup.sh" [New] 3L, 16B written
student@ubuntu:~$ chmod +x setup.sh
student@ubuntu:~$ ./setup.sh
student@ubuntu:~$
```

2. Define variable var1 and echo it

```
#
var1="Hello from Lab 6"
echo "var1: $var1"#
```

```
student@ubuntu:~$ ./setup.sh

var1: Hello from Lab 6#
student@ubuntu:~$
student@ubuntu:~$ _
```

3. Save output of ls -l into variable allFiles and echo it

```
#!/bin/bash
#
var1="Hello from Lab 6"
echo "var1: $var1"
Save ls -l to variable and display
allFiles="$(ls -l)"
echo "allFiles (ls -l):
 echo "$allFiles"
~
```

```
student@ubuntu:~$ ./setup.sh
var1: Hello from Lab 6
allFiles (ls -l):
total 8
drwxrwxr-x 2 student student 4096 Oct 30 14:57 dir1
-rwxr-x--- 1 student student    0 Oct 31 13:06 file1
-rwxrwxr-x 1 student student  199 Nov  1 11:55 setup.sh
student@ubuntu:~$ _
```

4. If directory dir1 exists echo a message; else create it

```bash
#!/bin/bash
#
## Define and show var1
#
var1="Hello from Lab 6"

echo "var1: $var1"
## Save ls -l to variable and display
allFiles="$(ls -l)"
echo "allFiles (ls -l):"
echo "$allFiles"
#drectory check

if [ -d "dir1" ]; then

          echo "Directory dir1 exists."

  else

            echo "Directory dir1 does not exist. Creating..."

              mkdir -p "dir1"

                echo "Directory dir1 created."

fi
```

```
setup.sh 28L, 393B written
student@ubuntu:~$ ./setup.sh
var1: Hello from Lab 6
allFiles (ls -l):
total 8
drwxrwxr-x 2 student student 4096 Oct 30 14:57 dir1
-rwxr-x--- 1 student student    0 Oct 31 13:06 file1
-rwxrwxr-x 1 student student  393 Nov  1 11:59 setup.sh
Directory dir1 exists.
student@ubuntu:~$
```

6. Check read, write, execute permissions on dir1/file2; grant missing user perms and show final ls

```bash
fi
# Permission checks for dir1/file2 (user permissions)

 f="dir1/file2"
 if [ ! -r "$f" ]; then
   echo "Read permission missing; granting to user..."
     chmod u+r "$f"
     fi

     if [ ! -w "$f" ]; then
      echo "Write permission missing; granting to user..."
         chmod u+w "$f"
         fi

         if [ ! -x "$f" ]; then

           echo "Execute permission missing; granting to user..."

          chmod u+x "$f"
             fi
             echo "Final permissions for $f:"
             ls -l "$f"
~
```

```
student@ubuntu:~$ ./setup.sh
var1: Hello from Lab 6
allFiles (ls -l):
total 8
drwxrwxr-x 2 student student 4096 Oct 30 14:57 dir1
-rwxr-x--- 1 student student    0 Oct 31 13:06 file1
-rwxrwxr-x 1 student student  906 Nov  1 12:04 setup.sh
Directory dir1 exists.
Execute permission missing; granting to user...
Final permissions for dir1/file2:
-rwxrw-r-- 1 student student 0 Oct 31 13:06 dir1/file2
student@ubuntu:~$ _
```

TASK NO 11:

Script setup.sh – argument comparisons (eq, ne, gt, lt, ge, le) and string checks

1. create file with shebang and set num and str variables

```
#!/bin/bash
num=$1

str=$2
```

```
student@ubuntu:~$ chmod +x setup.sh
student@ubuntu:~$ ./setup.sh 10 Student
```

2. add the -eq test (equal)

```
#!/bin/bash
num=$1

str=$2
if [ "$num" -eq 10 ]; then

          echo "$num is equal to 10 (-eq)."

  else

            echo "$num is NOT equal to 10 (-eq)."

fi
```

```
"setup.sh" 15L, 150B written
student@ubuntu:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
student@ubuntu:~$ ./setup.sh 7 Student
7 is NOT equal to 10 (-eq).
student@ubuntu:~$
```

3. add the -ne test (not equal)

```
#!/bin/bash
num=$1

str=$2
if [ "$num" -eq 10 ]; then

         echo "$num is equal to 10 (-eq)."

  else

           echo "$num is NOT equal to 10 (-eq)."

fi

if [ "$num" -ne 10 ]; then

         echo "$num is not equal to 10 (-ne)."

  else

           echo "$num is equal to 10 (-ne false)."

fi
```

```
"setup.sh" 25L, 278B written
student@ubuntu:~$ ./setup.sh 7 Student
7 is NOT equal to 10 (-eq).
7 is not equal to 10 (-ne).
student@ubuntu:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
student@ubuntu:~$ _
```

4. add the -gt test (greater than)

```
#!/bin/bash
num=$1

str=$2
if [ "$num" -eq 10 ]; then

          echo "$num is equal to 10 (-eq)."

  else

            echo "$num is NOT equal to 10 (-eq)."

fi

if [ "$num" -ne 10 ]; then

          echo "$num is not equal to 10 (-ne)."

  else

            echo "$num is equal to 10 (-ne false)."

fi

if [ "$num" -gt 10 ]; then

          echo "$num is greater than 10 (-gt)."

  else

            echo "$num is NOT greater than 10 (-gt)."

fi
```

```
"setup.sh" 35L, 408B written
student@ubuntu:~$ ./setup.sh 12 Student
12 is NOT equal to 10 (-eq).
12 is not equal to 10 (-ne).
12 is greater than 10 (-gt).
student@ubuntu:~$ ./setup.sh 9 Student
9 is NOT equal to 10 (-eq).
9 is not equal to 10 (-ne).
9 is NOT greater than 10 (-gt).
student@ubuntu:~$ _
```

5. add the -lt test (less than)

```
if [ "$num" -lt 10 ]; then

          echo "$num is less than 10 (-lt)."

  else

            echo "$num is NOT less than 10 (-lt)."

fi
```

```
"setup.sh" 45L, 532B written
student@ubuntu:~$ ./setup.sh 5 Student
5 is NOT equal to 10 (-eq).
5 is not equal to 10 (-ne).
5 is NOT greater than 10 (-gt).
5 is less than 10 (-lt).
student@ubuntu:~$ ./setup.sh 11 Student
11 is NOT equal to 10 (-eq).
11 is not equal to 10 (-ne).
11 is greater than 10 (-gt).
11 is NOT less than 10 (-lt).
student@ubuntu:~$
student@ubuntu:~$
```

6. add the -ge test (greater than or equal)

```
if [ "$num" -ge 10 ]; then
        echo "$num is greater than or equal to 10 (-ge)."
  else
          echo "$num is NOT greater than or equal to 10 (-ge)."
fi
```

```
"setup.sh" 55L, 686B written
student@ubuntu:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
student@ubuntu:~$ ./setup.sh 8 Student
8 is NOT equal to 10 (-eq).
8 is not equal to 10 (-ne).
8 is NOT greater than 10 (-gt).
8 is less than 10 (-lt).
8 is NOT greater than or equal to 10 (-ge).
student@ubuntu:~$ _
```

7. add the -le test (less than or equal)

```
if [ "$num" -le 10 ]; then
        echo "$num is less than or equal to 10 (-le)."
  else
          echo "$num is NOT less than or equal to 10 (-le)."
fi
```

```
"setup.sh" 65L, 834B written
student@ubuntu:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
student@ubuntu:~$ ./setup.sh 12 Student
12 is NOT equal to 10 (-eq).
12 is not equal to 10 (-ne).
12 is greater than 10 (-gt).
12 is NOT less than 10 (-lt).
12 is greater than or equal to 10 (-ge).
12 is NOT less than or equal to 10 (-le).
student@ubuntu:~$
```

8. string equality test (=)

```
if [ "$str" = "Student" ]; then

  echo "Second argument equals 'Student' ( = )."

else

        echo "Second argument does NOT equal 'Student' ( = )."

        fi
```

```
student@ubuntu:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
Second argument equals 'Student' ( = ).
student@ubuntu:~$ ./setup.sh 10 Test
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
Second argument does NOT equal 'Student' ( = ).
student@ubuntu:~$ _
```

9. string inequality test (!=)

```
if [ "$str" != "Student" ]; then

        echo "Second argument is not equal to 'Student' ( != )."

  else

          echo "Second argument equals 'Student' ( != false)."

fi
```

```
student@ubuntu:~$ ./setup.sh 10 Test
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
Second argument does NOT equal 'Student' ( = ).
Second argument is not equal to 'Student' ( != ).
student@ubuntu:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
Second argument equals 'Student' ( = ).
Second argument equals 'Student' ( != false).
student@ubuntu:~$
```

10. check if second argument is empty (zero-length)

```
if [ -z "$str" ]; then
         echo "Second argument is empty (zero-length)."
  else
           echo "Second argument is not empty."
fi
```

```
student@ubuntu:~$ ./setup.sh 10
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
Second argument does NOT equal 'Student' ( = ).
Second argument is not equal to 'Student' ( != ).
Second argument is empty (zero-length).
student@ubuntu:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
Second argument equals 'Student' ( = ).
Second argument equals 'Student' ( != false).
Second argument is not empty.
student@ubuntu:~$ _
```

TASK NO 12:

Script setup.sh – print all arguments with a for loop

1. Create the script with shebang and basic structure

```
#!/bin/bash
#
## Script to demonstrate printing all user-entered arguments using $*
#
#
```

```
"setup1.sh" [New] 5L, 88B written
student@ubuntu:~$ chmod +x setup1.sh
student@ubuntu:~$ ./setup1.sh
student@ubuntu:~$
```

2. Append the for loop using $* and print each argument

```
#!/bin/bash
#
## Script to demonstrate printing all user-entered arguments using $*
#
# Print all arguments using $*
#
echo "Printing all arguments using \$*:"

 for arg in $*; do

   echo "Argument: $arg"

   done
```

```
"setup1.sh" 14L, 217B written
student@ubuntu:~$ ./setup1.sh one "two words" three
Printing all arguments using $*:
Argument: one
Argument: two
Argument: words
Argument: three
student@ubuntu:~$
```

TASK NO 13:

Script setup.sh – while loop summation and functions

1. Add the shebang line

```
#!/bin/bash
#
```

```
"setup.sh" 3L, 16B written
student@ubuntu:~$ chmod +x setup.sh
student@ubuntu:~$ ./setup.sh
student@ubuntu:~$
```

2. Add the while-loop summation (interactive)

```
#!/bin/bash
#
# While-loop summation (interactive)

_sum=0
 while true; do

   read -p "Enter a number (or 'q' to quit): " input

     if [ "$input" = "q" ]; then

         break

           fi
             sum=$((sum + input))

               echo "Total Score: $sum"
               done
                echo "Final total: $sum"
```

```
student@ubuntu:~$ ./setup.sh
Enter a number (or 'q' to quit): 2
Total Score: 2
Enter a number (or 'q' to quit): 5
Total Score: 7
Enter a number (or 'q' to quit): 4
Total Score: 11
Enter a number (or 'q' to quit): q
Final total: 11
student@ubuntu:~$
```

3. Add the interactive summation function and demonstrate it

```
# Function to accumulate scores interactively

 sum_two() {

   sum=0

     while true; do

         read -p "Enter a number (or 'q' to quit): " input

             if [ "$input" = "q" ]; then

                   break

                       fi

                           sum=$((sum + input))

                               echo "Total Score: $sum"

                                 done

                                   echo "Function final total: $sum"

                                   }


                                   # Demonstrate the function

                                   echo "Now calling sum_two function:"

                                   sum_two
```

```
student@ubuntu:~$ ./setup.sh
Enter a number (or 'q' to quit): 3
Total Score: 3
Enter a number (or 'q' to quit): 4
Total Score: 7
Enter a number (or 'q' to quit): 5
Total Score: 12
Enter a number (or 'q' to quit): q
Final total: 12
Now calling sum_two function:
Enter a number (or 'q' to quit): _
```

4. Add a function that takes two numeric arguments, sums them, and returns the result

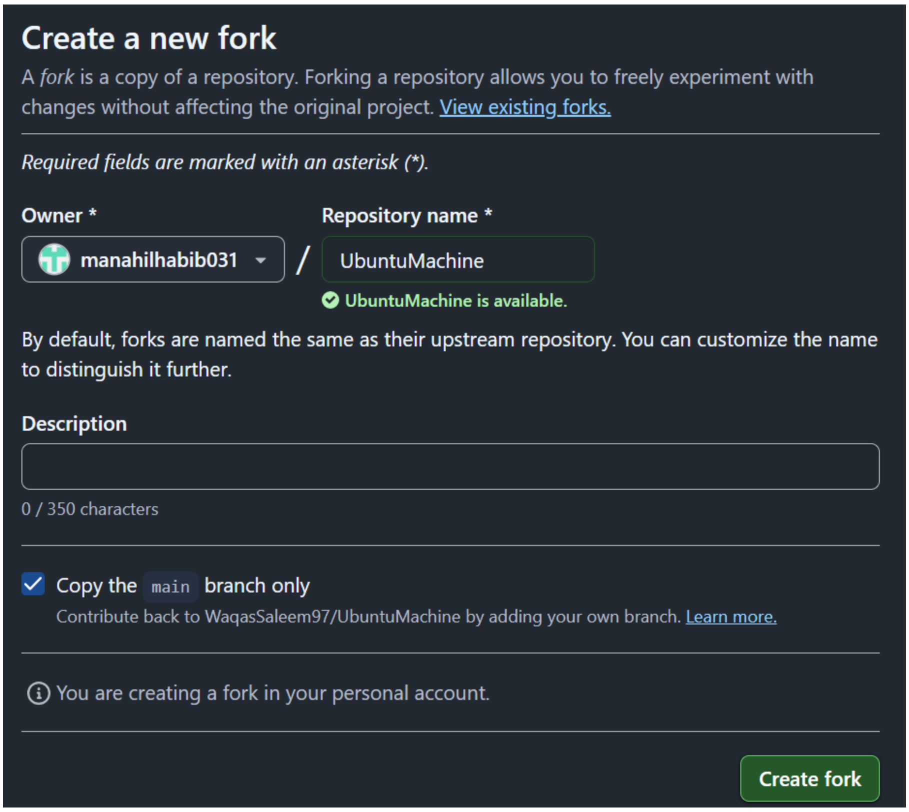
```
# Function that sums two arguments and returns the result
#
_sum_args() {

   a=$1

     b=$2

       return $((a + b))

      }

       # Demonstrate sum_args function

       echo "Now demonstrating sum_args function:"

       sum_args 3 4

       result=$?
       echo "sum_args(3,4) returned: $result"
~
~
~
```

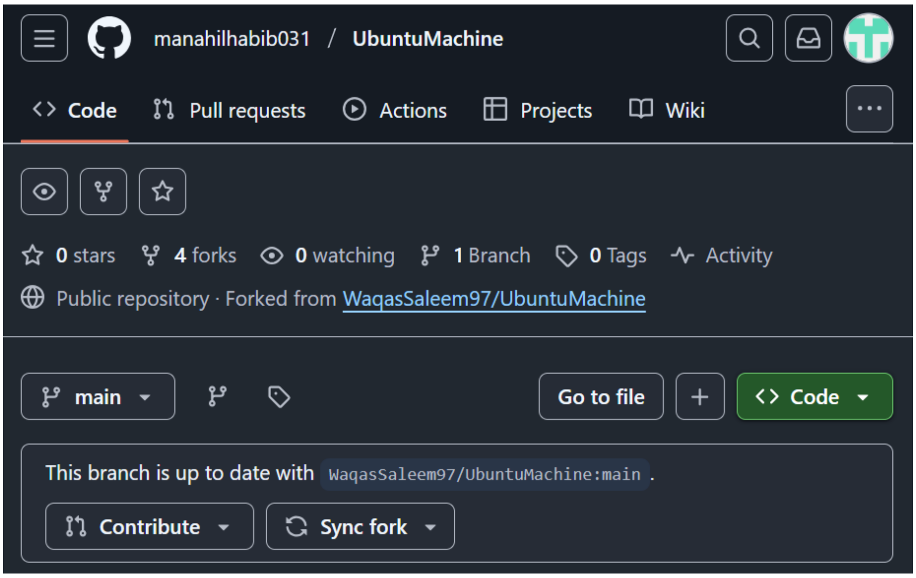
```
"setup.sh" 74L, 1369B written
student@ubuntu:~$ chmod +x setup.sh
student@ubuntu:~$ ./setup.sh
Enter a number (or 'q' to quit): 1
Total Score: 1
Enter a number (or 'q' to quit): 2
Total Score: 3
Enter a number (or 'q' to quit): q
Final total: 3
Now calling sum_two function:
Enter a number (or 'q' to quit): q
Function final total: 0
Now demonstrating sum_args function:
sum_args(3,4) returned: 7
student@ubuntu:~$ _
```

TASK NO 14:

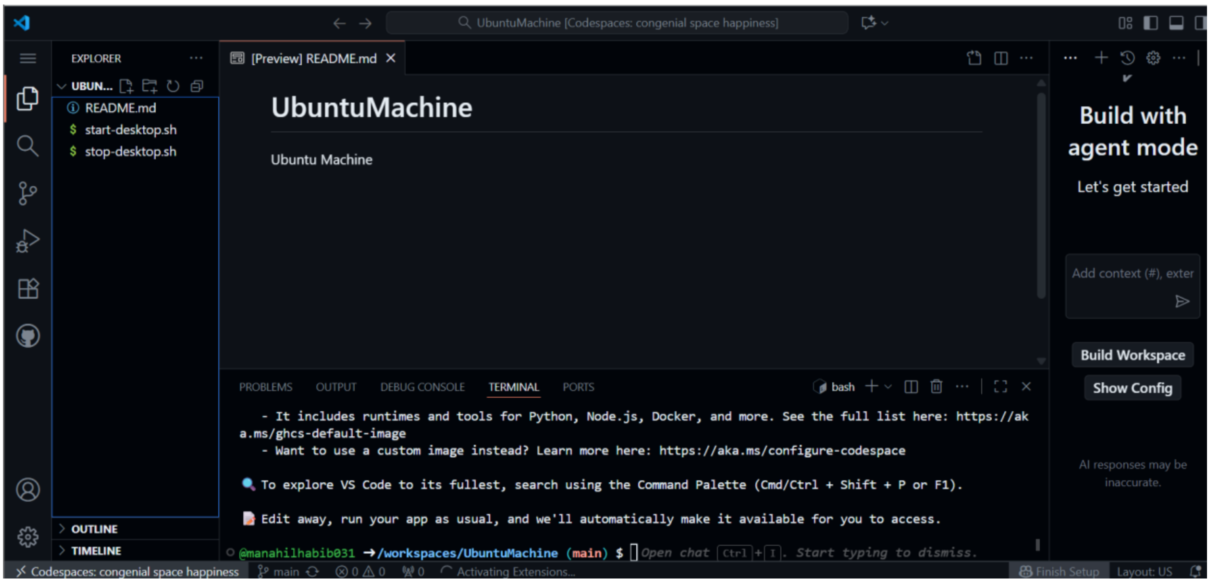
Codespaces GUI — fork repo, run start-desktop.sh, open VNC, stop GUI

1. Fork the repository to your GitHub account

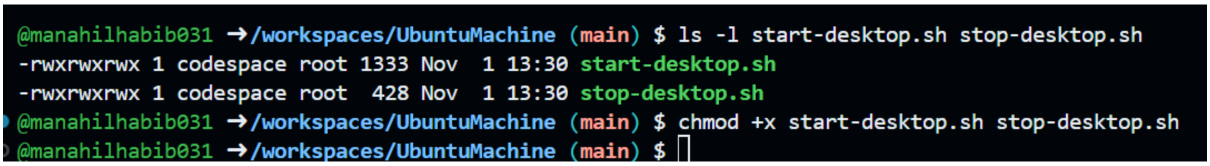




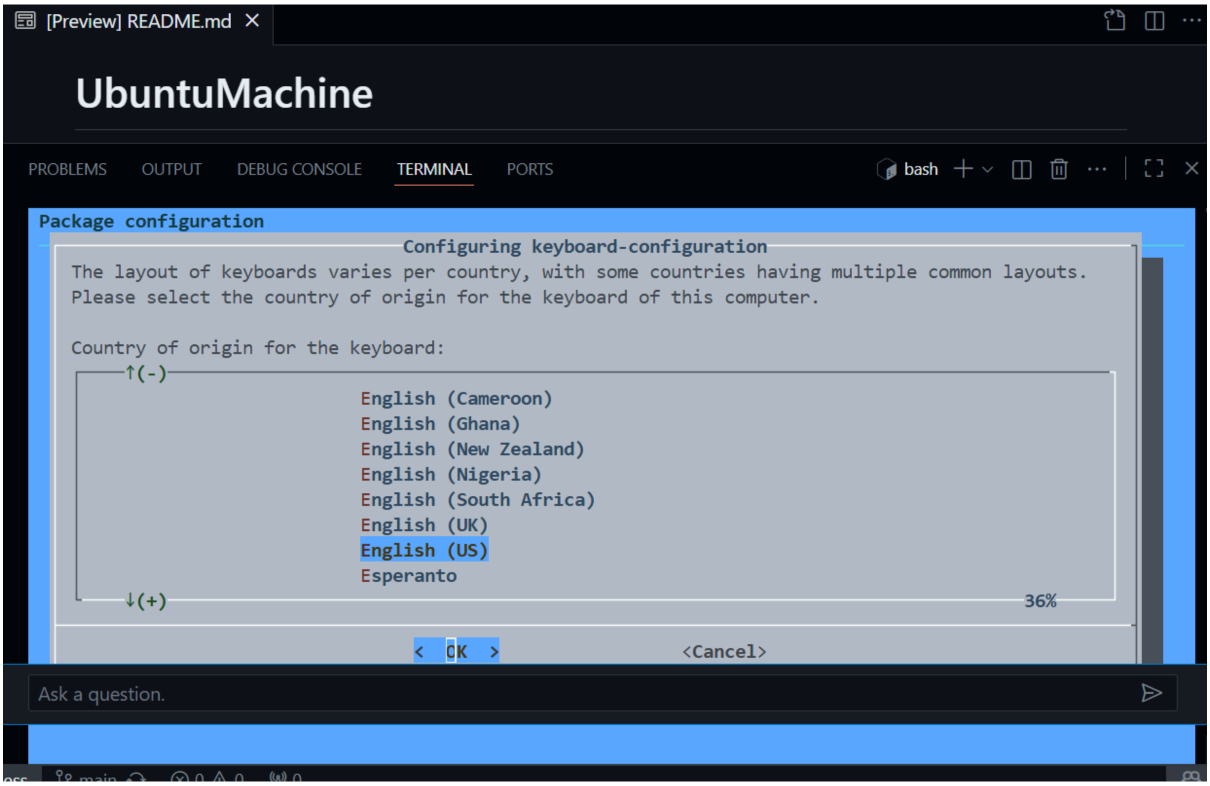
2. Open a Codespace on your fork



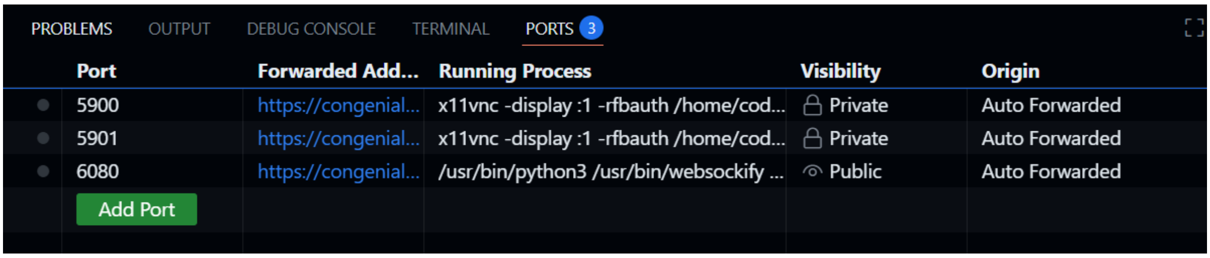
3. Verify the start script is present and executable (capture evidence)



4. Run the start script inside the Codespace terminal

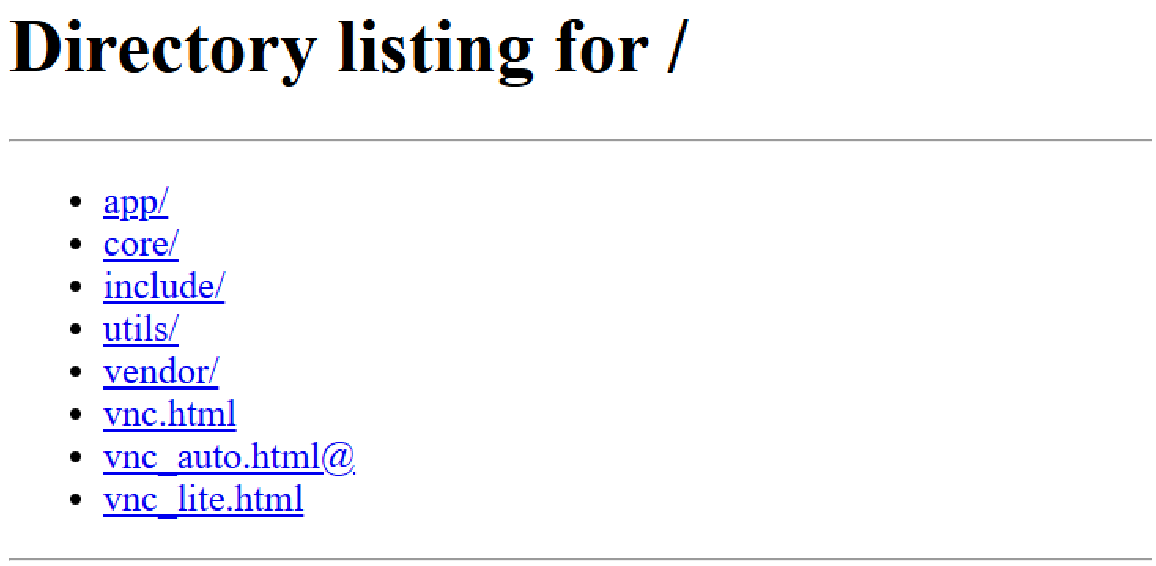


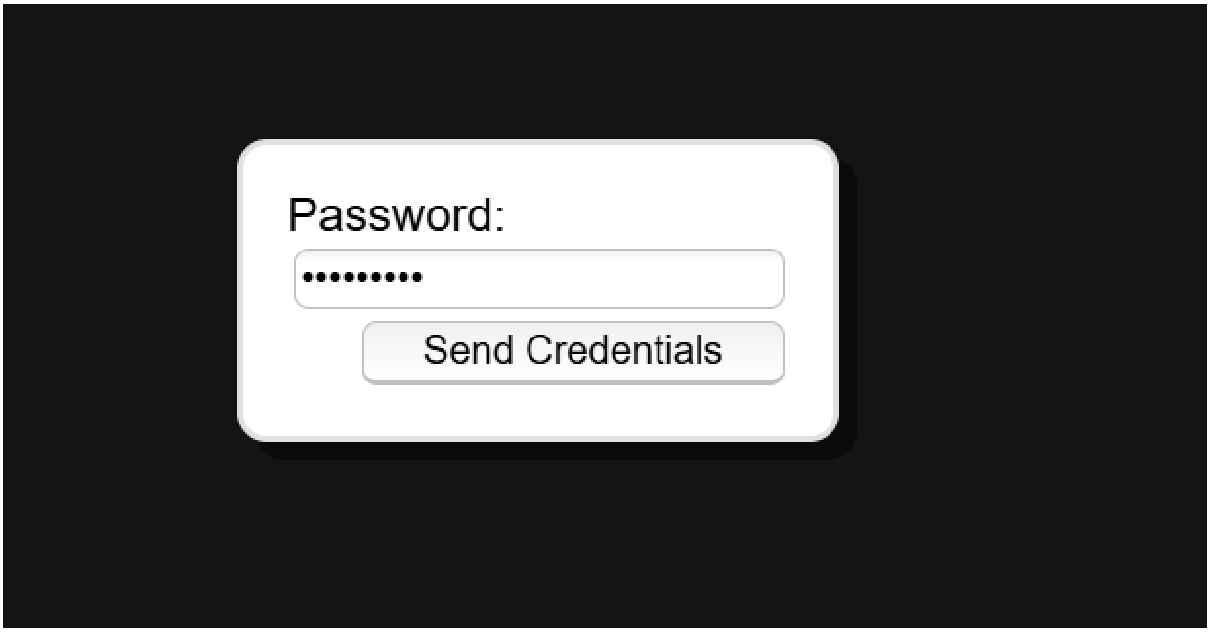
5. Verify forwarded ports in Codespaces (Ports view)

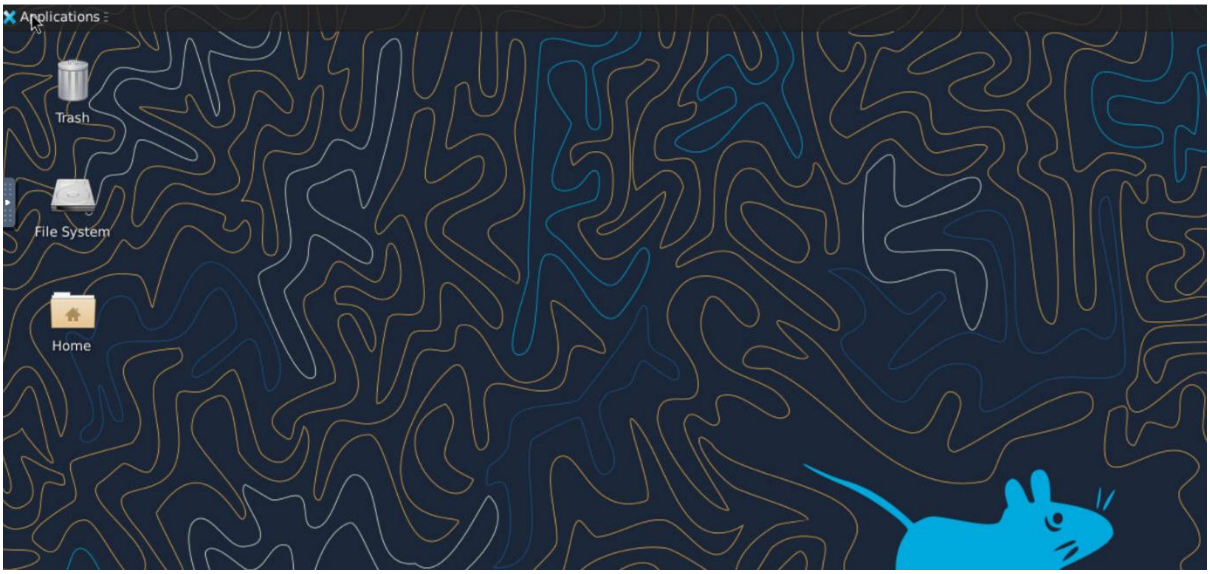


Port	Forwarded Add...	Running Process	Visibility	Origin
5900	https://congenial...	x11vnc -display :1 -rfbauth /home/cod...	Private	Auto Forwarded
5901	https://congenial...	x11vnc -display :1 -rfbauth /home/cod...	Private	Auto Forwarded
6080	https://congenial...	/usr/bin/python3 /usr/bin/websockify ...	Public	Auto Forwarded

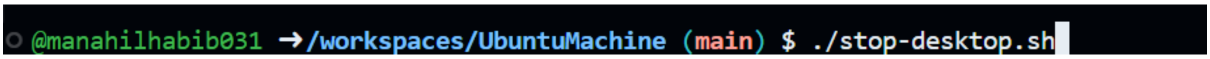
6. Open forwarded port 6080 and connect to VNC HTML page







7. Stop the GUI



Exam Evaluation Questions

1. Group Management and Membership

Scenario:

Create groups and manage a user’s primary and supplementary group memberships.

Steps:

1. Create groups g1, g2, and g3.

```
manahil@ubuntu:~$ sudo groupadd g1
[sudo] password for manahil:
manahil@ubuntu:~$ sudo groupadd g2
manahil@ubuntu:~$ sudo groupadd g3
```

2. Change examuser's primary group to g3 and add g1 and g2 as supplementary groups.

```
manahil@ubuntu:~$ sudo id examuser || sudo adduser --disabled-password --gecos "" examuser
id: 'examuser': no such user
info: Adding user `examuser' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `examuser' (1007) ...
info: Adding new user `examuser' (1007) with group `examuser (1007)' ...
info: Creating home directory `/home/examuser' ...
info: Copying files from `/etc/skel' ...
info: Adding new user `examuser' to supplemental / extra groups `users' ...
info: Adding user `examuser' to group `users' ...
manahil@ubuntu:~$ sudo usermod -g g3 examuser
manahil@ubuntu:~$ sudo usermod -aG g1,g2 examuser
manahil@ubuntu:~$ _
```

3. Show the final id and /etc/group lines that prove the changes.

```
manahil@ubuntu:~$ id examuser
uid=1007(examuser) gid=1013(g3) groups=1013(g3),100(users),1011(g1),1012(g2)
manahil@ubuntu:~$ grep -E '(^g1:|^g2:|^g3:)' /etc/group
g1:x:1011:examuser
g2:x:1012:examuser
g3:x:1013:
manahil@ubuntu:~$
```

---

2. Ownership and Permission Tasks

Scenario:

Demonstrate ownership changes and apply both symbolic and numeric permission changes.

Steps:

1. Create workspace/secret.txt, change its owner to examuser and group to g1.

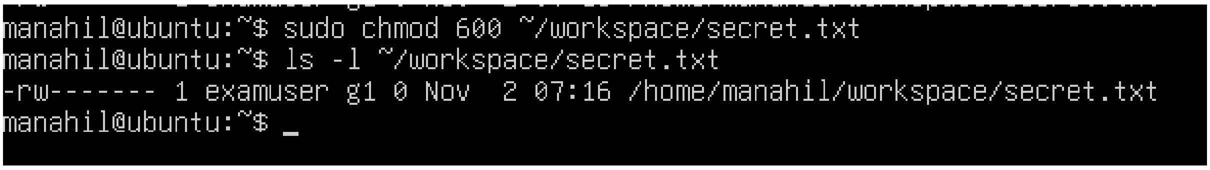
```
manahil@ubuntu:~$ mkdir -p ~/workspace
manahil@ubuntu:~$ touch ~/workspace/secret.txt
manahil@ubuntu:~$
```

2. Remove all permissions for group and others using a symbolic command, then using a numeric command to achieve the same result.

```
manahil@ubuntu:~$ sudo chown examuser:g1 ~/workspace/secret.txt
manahil@ubuntu:~$ ls -l ~/workspace/secret.txt
-rw-rw-r-- 1 examuser g1 0 Nov  2 07:16 /home/manahil/workspace/secret.txt
manahil@ubuntu:~$
```

3. Show ls -l for the file after each change to document the permission bits.

```
manahil@ubuntu:~$ sudo chmod go-rwx ~/workspace/secret.txt
manahil@ubuntu:~$ ls -l ~/workspace/secret.txt
-rw------- 1 examuser g1 0 Nov  2 07:16 /home/manahil/workspace/secret.txt
manahil@ubuntu:~$ _
```



```
manahil@ubuntu:~$ sudo chmod 600 ~/workspace/secret.txt
manahil@ubuntu:~$ ls -l ~/workspace/secret.txt
-rw------- 1 examuser g1 0 Nov  2 07:16 /home/manahil/workspace/secret.txt
manahil@ubuntu:~$ _
```

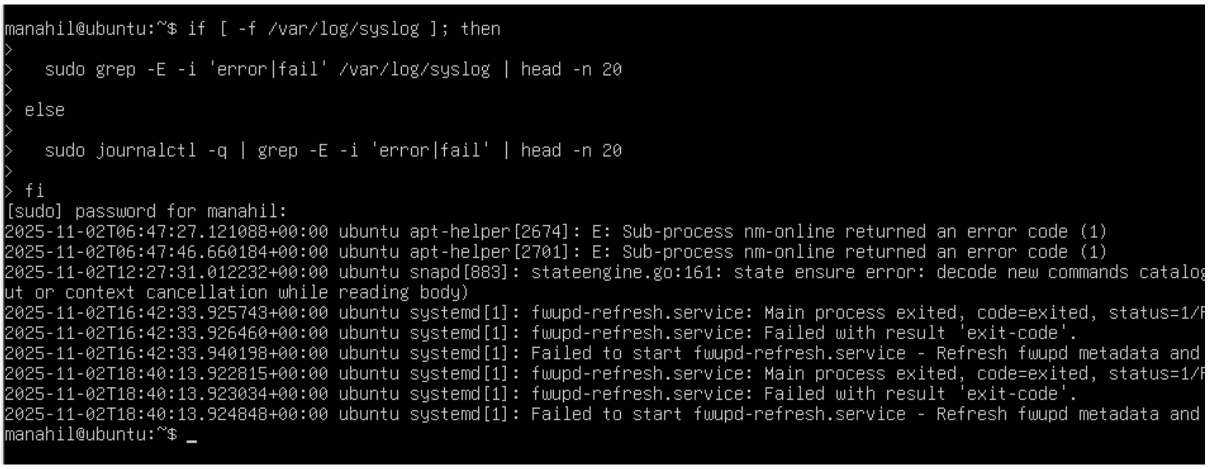
3. Pipes, Grep, and Redirection Practice

Scenario:

Filter system logs and save results using redirection and piping.

Steps:

1. Use grep (or journalctl where applicable) with a pipe to find lines containing "error" or "fail" and show the first 20 results.



```
manahil@ubuntu:~$ if [ -f /var/log/syslog ]; then
>
>   sudo grep -E -i 'error|fail' /var/log/syslog | head -n 20
>
> else
>
>   sudo journalctl -q | grep -E -i 'error|fail' | head -n 20
>
> fi
[sudo] password for manahil:
2025-11-02T06:47:27.121088+00:00 ubuntu apt-helper[2674]: E: Sub-process nm-online returned an error code (1)
2025-11-02T06:47:46.660184+00:00 ubuntu apt-helper[2701]: E: Sub-process nm-online returned an error code (1)
2025-11-02T12:27:31.012232+00:00 ubuntu snapd[883]: stateengine.go:161: state ensure error: decode new commands catalog
ut or context cancellation while reading body)
2025-11-02T16:42:33.925743+00:00 ubuntu systemd[1]: fwupd-refresh.service: Main process exited, code=exited, status=1/F
2025-11-02T16:42:33.926460+00:00 ubuntu systemd[1]: fwupd-refresh.service: Failed with result 'exit-code'.
2025-11-02T16:42:33.940198+00:00 ubuntu systemd[1]: Failed to start fwupd-refresh.service - Refresh fwupd metadata and
2025-11-02T18:40:13.922815+00:00 ubuntu systemd[1]: fwupd-refresh.service: Main process exited, code=exited, status=1/F
2025-11-02T18:40:13.923034+00:00 ubuntu systemd[1]: fwupd-refresh.service: Failed with result 'exit-code'.
2025-11-02T18:40:13.924848+00:00 ubuntu systemd[1]: Failed to start fwupd-refresh.service - Refresh fwupd metadata and
manahil@ubuntu:~$ _
```

2. Save the filtered results to a file ~/logs/errors.txt using overwrite, then append additional matching lines using append redirection.

```
manahil@ubuntu:~$ mkdir -p ~/logs

imanahil@ubuntu:~$
manahil@ubuntu:~$ if [ -f /var/log/syslog ]; then
>
>   sudo grep -E -i 'error|fail' /var/log/syslog > ~/logs/errors.txt
>
> else
>
>   sudo journalctl -q | grep -E -i 'error|fail' > ~/logs/errors.txt
>
> fi
manahil@ubuntu:~$ ls -l ~/logs/errors.txt
-rw-rw-r-- 1 manahil manahil 1157 Nov  2 18:44 /home/manahil/logs/errors.txt
manahil@ubuntu:~$
```

3. Use a pager to view the saved file.

```
manahil@ubuntu:~$ ls -l ~/logs/errors.txt
-rw-rw-r-- 1 manahil manahil 1157 Nov  2 18:44 /home/manahil/logs/errors.txt
manahil@ubuntu:~$ if [ -f /var/log/syslog ]; then
>
>   sudo grep -E -i 'warning|fail' /var/log/syslog >> ~/logs/errors.txt
>
> else
>
>   sudo journalctl -q | grep -E -i 'warning|fail' >> ~/logs/errors.txt
>
> fi
#
# Show tmanahil@ubuntu:~$ he file content

cat ~/logs/errors.txt | head -n 50Command 'he' not found, but can be installed with:
sudo apt install node-he
manahil@ubuntu:~$
manahil@ubuntu:~$ cat ~/logs/errors.txt | head -n 50
2025-11-02T06:47:27.121088+00:00 ubuntu apt-helper[2674]: E: Sub-process nm-online returned an error code (1)
2025-11-02T06:47:46.660184+00:00 ubuntu apt-helper[2701]: E: Sub-process nm-online returned an error code (1)
2025-11-02T12:27:31.012232+00:00 ubuntu snapd[883]: stateengine.go:161: state ensure error: decode new commands catal
ut or context cancellation while reading body)
2025-11-02T16:42:33.925743+00:00 ubuntu systemd[1]: fwupd-refresh.service: Main process exited, code=exited, status=1
2025-11-02T16:42:33.926460+00:00 ubuntu systemd[1]: fwupd-refresh.service: Failed with result 'exit-code'.
2025-11-02T16:42:33.940198+00:00 ubuntu systemd[1]: Failed to start fwupd-refresh.service - Refresh fwupd metadata an
2025-11-02T18:40:13.922815+00:00 ubuntu systemd[1]: fwupd-refresh.service: Main process exited, code=exited, status=1
2025-11-02T18:40:13.923034+00:00 ubuntu systemd[1]: fwupd-refresh.service: Failed with result 'exit-code'.
2025-11-02T18:40:13.924848+00:00 ubuntu systemd[1]: Failed to start fwupd-refresh.service - Refresh fwupd metadata an
2025-11-02T16:42:33.925743+00:00 ubuntu systemd[1]: fwupd-refresh.service: Main process exited, code=exited, status=1
2025-11-02T16:42:33.926460+00:00 ubuntu systemd[1]: fwupd-refresh.service: Failed with result 'exit-code'.
2025-11-02T16:42:33.940198+00:00 ubuntu systemd[1]: Failed to start fwupd-refresh.service - Refresh fwupd metadata an
2025-11-02T18:40:13.922815+00:00 ubuntu systemd[1]: fwupd-refresh.service: Main process exited, code=exited, status=1
2025-11-02T18:40:13.923034+00:00 ubuntu systemd[1]: fwupd-refresh.service: Failed with result 'exit-code'.
2025-11-02T18:40:13.924848+00:00 ubuntu systemd[1]: Failed to start fwupd-refresh.service - Refresh fwupd metadata an
manahil@ubuntu:~$ _
```

```
manahil@ubuntu:~$ cat ~/logs/errors.txt | head -n 50
2025-11-02T06:47:27.121088+00:00 ubuntu apt-helper[2674]: E: Sub-process nm-online returned an error code (1)
2025-11-02T06:47:46.660184+00:00 ubuntu apt-helper[2701]: E: Sub-process nm-online returned an error code (1)
2025-11-02T12:27:31.012232+00:00 ubuntu snapd[883]: stateengine.go:161: state ensure error: decode new commands catalog
ut or context cancellation while reading body)
2025-11-02T16:42:33.925743+00:00 ubuntu systemd[1]: fwupd-refresh.service: Main process exited, code=exited, status=1/F
2025-11-02T16:42:33.926460+00:00 ubuntu systemd[1]: fwupd-refresh.service: Failed with result 'exit-code'.
2025-11-02T16:42:33.940198+00:00 ubuntu systemd[1]: Failed to start fwupd-refresh.service - Refresh fwupd metadata and
2025-11-02T18:40:13.922815+00:00 ubuntu systemd[1]: fwupd-refresh.service: Main process exited, code=exited, status=1/F
2025-11-02T18:40:13.923034+00:00 ubuntu systemd[1]: fwupd-refresh.service: Failed with result 'exit-code'.
2025-11-02T18:40:13.924848+00:00 ubuntu systemd[1]: Failed to start fwupd-refresh.service - Refresh fwupd metadata and
2025-11-02T16:42:33.925743+00:00 ubuntu systemd[1]: fwupd-refresh.service: Main process exited, code=exited, status=1/F
2025-11-02T16:42:33.926460+00:00 ubuntu systemd[1]: fwupd-refresh.service: Failed with result 'exit-code'.
2025-11-02T16:42:33.940198+00:00 ubuntu systemd[1]: Failed to start fwupd-refresh.service - Refresh fwupd metadata and
2025-11-02T18:40:13.922815+00:00 ubuntu systemd[1]: fwupd-refresh.service: Main process exited, code=exited, status=1/F
2025-11-02T18:40:13.923034+00:00 ubuntu systemd[1]: fwupd-refresh.service: Failed with result 'exit-code'.
2025-11-02T18:40:13.924848+00:00 ubuntu systemd[1]: Failed to start fwupd-refresh.service - Refresh fwupd metadata and
manahil@ubuntu:~$ less ~/logs/errors.txt
2025-11-02T06:47:27.121088+00:00 ubuntu apt-helper[2674]: E: Sub-process nm-online returned an error code (1)
2025-11-02T06:47:46.660184+00:00 ubuntu apt-helper[2701]: E: Sub-process nm-online returned an error code (1)
2025-11-02T12:27:31.012232+00:00 ubuntu snapd[883]: stateengine.go:161: state ensure error: decode new commands catalog
ut or context cancellation while reading body)
2025-11-02T16:42:33.925743+00:00 ubuntu systemd[1]: fwupd-refresh.service: Main process exited, code=exited, status=1/F
2025-11-02T16:42:33.926460+00:00 ubuntu systemd[1]: fwupd-refresh.service: Failed with result 'exit-code'.
2025-11-02T16:42:33.940198+00:00 ubuntu systemd[1]: Failed to start fwupd-refresh.service - Refresh fwupd metadata and
2025-11-02T18:40:13.922815+00:00 ubuntu systemd[1]: fwupd-refresh.service: Main process exited, code=exited, status=1/F
2025-11-02T18:40:13.923034+00:00 ubuntu systemd[1]: fwupd-refresh.service: Failed with result 'exit-code'.
2025-11-02T18:40:13.924848+00:00 ubuntu systemd[1]: Failed to start fwupd-refresh.service - Refresh fwupd metadata and
2025-11-02T16:42:33.925743+00:00 ubuntu systemd[1]: fwupd-refresh.service: Main process exited, code=exited, status=1/F
2025-11-02T16:42:33.926460+00:00 ubuntu systemd[1]: fwupd-refresh.service: Failed with result 'exit-code'.
2025-11-02T16:42:33.940198+00:00 ubuntu systemd[1]: Failed to start fwupd-refresh.service - Refresh fwupd metadata and
2025-11-02T18:40:13.922815+00:00 ubuntu systemd[1]: fwupd-refresh.service: Main process exited, code=exited, status=1/F
2025-11-02T18:40:13.923034+00:00 ubuntu systemd[1]: fwupd-refresh.service: Failed with result 'exit-code'.
2025-11-02T18:40:13.924848+00:00 ubuntu systemd[1]: Failed to start fwupd-refresh.service - Refresh fwupd metadata and
/home/manahil/logs/errors.txt (END)
```

---

4. Script: Variables, Command Substitution, File & Dir Checks

Scenario:

Build and run a script incrementally that demonstrates variables, command substitution, and filesystem checks.

Steps:

1. Create setup.sh with a shebang and a variable var1 that you echo.

```
#!/bin/bash
#
## Define and show var1
#
var1="Hello from lab6"
#
echo "var1: $var1"
~
```

```
manahil@ubuntu:~$ chmod +x setup.sh
manahil@ubuntu:~$ ./setup.sh
var1: Hello from lab6
manahil@ubuntu:~$
```

2. Append command substitution that stores ls -l output into a variable and echo it.

```
#!/bin/bash
#
## Define and show var1
#
var1="Hello from lab6"
#
echo "var1: $var1"
allFiles="$(ls -l)"

echo "allFiles (ls -l):"

echo "$allFiles"
~
```

```
manahil@ubuntu:~$ ./setup.sh
var1: Hello from lab6
allFiles (ls -l):
total 352404
drwxrwxr-x 4 manahil manahil      4096 Oct 19 01:43 analusis
-rw-rw-r-- 1 manahil manahil       426 Oct 17 04:52 answers.md
-rw-rw-r-- 1 manahil manahil       254 Oct 24 07:11 apt_update_output.txt
-rw-rw-r-- 1 manahil manahil       418 Oct 24 00:17 apt_update_vsupgrade.md
drwxr-xr-x 2 manahil manahil      4096 Oct 24 04:55 Desktop
drwxr-xr-x 2 manahil manahil      4096 Oct 24 04:55 Documents
drwxr-xr-x 2 manahil manahil      4096 Oct 24 04:55 Downloads
-rw-rw-r-- 1 manahil manahil 120234260 Oct 21 20:35 google-chrome-stable_current_amd64.deb
-rw-rw-r-- 1 manahil manahil 120234260 Oct 21 20:35 google-chrome-stable_current_amd64.deb.1
-rw-rw-r-- 1 manahil manahil 120234260 Oct 21 20:35 google-chrome-stable_current_amd64.deb.2
-rw-rw-r-- 1 manahil manahil       296 Oct 24 09:51 k8s-sample.yaml
drwxrwxr-x 3 manahil manahil      4096 Oct 17 04:53 lab4
drwxrwxr-x 2 manahil manahil      4096 Oct 24 10:28 Lab5
drwxrwxr-x 2 manahil manahil      4096 Nov  2 18:44 logs
drwxr-xr-x 2 manahil manahil      4096 Oct 24 04:55 Music
drwxr-xr-x 2 manahil manahil      4096 Oct 24 04:55 Pictures
drwxr-xr-x 2 manahil manahil      4096 Oct 24 04:55 Public
-rw-rw-r-- 1 manahil manahil        20 Sep 27 12:19 README.md
-rwxrwxr-x 1 manahil manahil       103 Oct 19 01:13 run-demo.sh
-rwxrwxr-x 1 manahil manahil       148 Nov  2 18:50 setup.sh
drwx------ 3 manahil manahil      4096 Oct 24 04:15 snap
-rw-rw-r-- 1 manahil docker      48072 Nov  1 08:06 syslog_systemd.txt
drwxr-xr-x 2 manahil manahil      4096 Oct 24 04:55 Templates
drwxrwxr-t 2 manahil manahil      4096 Oct 24 04:55 thinclient_drives
drwxr-xr-x 2 manahil manahil      4096 Oct 24 04:55 Videos
drwxrwxr-x 2 manahil manahil      4096 Nov  2 07:16 workspace
manahil@ubuntu:~$
```

3. Append directory and file checks that create dir1 and dir1/file2 if missing, and display their final permissions.

```
echo "$allFiles"
# Directory check
#
 if [ -d "dir1" ]; then

   echo "Directory dir1 exists."

   else

     echo "Directory dir1 does not exist. Creating..."

       mkdir -p "dir1"

         echo "Directory dir1 created."
         fi

#         # File check

         if [ -f "dir1/file2" ]; then

           echo "file2 already exists."

           else

             echo "file2 does not exist. Creating..."

               touch "dir1/file2"

                 chmod a-rwx "dir1/file2"
                   echo "file2 created."
                   fi
#                   # Final listing
#

                   ls -ld dir1 dir1/file2
~
~
~
```

```
manahil@ubuntu:~$ ./setup.sh
var1: Hello from lab6
allFiles (ls -l):
total 352404
drwxrwxr-x 4 manahil manahil      4096 Oct 19 01:43 analusis
-rw-rw-r-- 1 manahil manahil       426 Oct 17 04:52 answers.md
-rw-rw-r-- 1 manahil manahil       254 Oct 24 07:11 apt_update_output.txt
-rw-rw-r-- 1 manahil manahil       418 Oct 24 00:17 apt_update_vsupgrade.md
drwxr-xr-x 2 manahil manahil      4096 Oct 24 04:55 Desktop
drwxr-xr-x 2 manahil manahil      4096 Oct 24 04:55 Documents
drwxr-xr-x 2 manahil manahil      4096 Oct 24 04:55 Downloads
-rw-rw-r-- 1 manahil manahil 120234260 Oct 21 20:35 google-chrome-stable_current_amd64.deb
-rw-rw-r-- 1 manahil manahil 120234260 Oct 21 20:35 google-chrome-stable_current_amd64.deb.1
-rw-rw-r-- 1 manahil manahil 120234260 Oct 21 20:35 google-chrome-stable_current_amd64.deb.2
-rw-rw-r-- 1 manahil manahil       296 Oct 24 09:51 k8s-sample.yaml
drwxrwxr-x 3 manahil manahil      4096 Oct 17 04:53 lab4
drwxrwxr-x 2 manahil manahil      4096 Oct 24 10:28 Lab5
drwxrwxr-x 2 manahil manahil      4096 Nov  2 18:44 logs
drwxr-xr-x 2 manahil manahil      4096 Oct 24 04:55 Music
drwxr-xr-x 2 manahil manahil      4096 Oct 24 04:55 Pictures
drwxr-xr-x 2 manahil manahil      4096 Oct 24 04:55 Public
-rw-rw-r-- 1 manahil manahil        20 Sep 27 12:19 README.md
-rwxrwxr-x 1 manahil manahil       103 Oct 19 01:13 run-demo.sh
-rwxrwxr-x 1 manahil manahil       765 Nov  2 18:54 setup.sh
drwx------ 3 manahil manahil      4096 Oct 24 04:15 snap
-rw-rw-r-- 1 manahil docker      48072 Nov  1 08:06 syslog_systemd.txt
drwxr-xr-x 2 manahil manahil      4096 Oct 24 04:55 Templates
drwxrwxr-t 2 manahil manahil      4096 Oct 24 04:55 thinclient_drives
drwxr-xr-x 2 manahil manahil      4096 Oct 24 04:55 Videos
drwxrwxr-x 2 manahil manahil      4096 Nov  2 07:16 workspace
Directory dir1 does not exist. Creating...
Directory dir1 created.
file2 does not exist. Creating...
file2 created.
drwxrwxr-x 2 manahil manahil 4096 Nov  2 18:54 dir1
---------- 1 manahil manahil    0 Nov  2 18:54 dir1/file2
manahil@ubuntu:~$ _
```

---

5. Script: Comparisons and String Tests

Scenario:

Incrementally add numeric and string comparison tests to a script and show both true/false cases.

Steps:

1. Overwrite setup.sh to set num=$1 and str=$2, and add an -eq test showing true and false examples.

```
#!/bin/bash
#
num=$1
str=$2
~
```

```
"setup.sh" 4L, 28B written
manahil@ubuntu:~$ chmod +x setup.sh

./smanahil@ubuntu:~$
manahil@ubuntu:~$ ./setup.sh 10 Student
```

2. Append -ne, -gt, -lt, -ge, and -le tests and demonstrate at least one true and one false invocation for each.

```
#!/bin/bash
#
num=$1
str=$2
if [ "$num" -eq 10 ]; then

          echo "$num is equal to 10 (-eq)."

  else

            echo "$num is NOT equal to 10 (-eq)."

fi
```

```
manahil@ubuntu:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
manahil@ubuntu:~$ ./setup.sh 7 Student
7 is NOT equal to 10 (-eq).
manahil@ubuntu:~$ _
```

```
#!/bin/bash
#
num=$1
str=$2
if [ "$num" -eq 10 ]; then

          echo "$num is equal to 10 (-eq)."

  else

            echo "$num is NOT equal to 10 (-eq)."

fi
if [ "$num" -ne 10 ]; then

          echo "$num is not equal to 10 (-ne)."

  else

            echo "$num is equal to 10 (-ne false)."

fi
~
```

```
"setup.sh" 22L, 276B written
manahil@ubuntu:~$ ./setup.sh 7 Student
7 is NOT equal to 10 (-eq).
7 is not equal to 10 (-ne).
manahil@ubuntu:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
manahil@ubuntu:~$ _
```

```
if [ "$num" -gt 10 ]; then

          echo "$num is greater than 10 (-gt)."

  else

            echo "$num is NOT greater than 10 (-gt)."

fi_
```

```
"setup.sh" 31L, 405B written
manahil@ubuntu:~$ ^C
manahil@ubuntu:~$ ./setup.sh 12 Student
12 is NOT equal to 10 (-eq).
12 is not equal to 10 (-ne).
12 is greater than 10 (-gt).
manahil@ubuntu:~$ ^C
manahil@ubuntu:~$ ^C
manahil@ubuntu:~$ ./setup.sh 9 Student
9 is NOT equal to 10 (-eq).
9 is not equal to 10 (-ne).
9 is NOT greater than 10 (-gt).
manahil@ubuntu:~$
```

```
if [ "$num" -lt 10 ]; then

          echo "$num is less than 10 (-lt)."

  else

            echo "$num is NOT less than 10 (-lt)."

fi_
```

```
"setup.sh" 40L, 528B written
manahil@ubuntu:~$ ./setup.sh 5 Student
5 is NOT equal to 10 (-eq).
5 is not equal to 10 (-ne).
5 is NOT greater than 10 (-gt).
5 is less than 10 (-lt).
manahil@ubuntu:~$ ./setup.sh 11 Student
11 is NOT equal to 10 (-eq).
11 is not equal to 10 (-ne).
11 is greater than 10 (-gt).
11 is NOT less than 10 (-lt).
manahil@ubuntu:~$
```

```
if [ "$num" -ge 10 ]; then

        echo "$num is greater than or equal to 10 (-ge)."

  else

          echo "$num is NOT greater than or equal to 10 (-ge)."

fi_
```

```
manahil@ubuntu:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
manahil@ubuntu:~$ ./setup.sh 8 Student
8 is NOT equal to 10 (-eq).
8 is not equal to 10 (-ne).
8 is NOT greater than 10 (-gt).
8 is less than 10 (-lt).
8 is NOT greater than or equal to 10 (-ge).
manahil@ubuntu:~$
```

```
if [ "$num" -le 10 ]; then

        echo "$num is less than or equal to 10 (-le)."

  else

          echo "$num is NOT less than or equal to 10 (-le)."

fi
```

```
manahil@ubuntu:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
manahil@ubuntu:~$ ./setup.sh 12 Student
12 is NOT equal to 10 (-eq).
12 is not equal to 10 (-ne).
12 is greater than 10 (-gt).
12 is NOT less than 10 (-lt).
12 is greater than or equal to 10 (-ge).
12 is NOT less than or equal to 10 (-le).
manahil@ubuntu:~$ _
```

3. Append string equality (=) and inequality (!=) checks and a -z (zero-length) test for the second argument, demonstrating true/false cases.

```
if [ "$str" = "Student" ]; then

          echo "Second argument equals 'Student' ( = )."

  else

            echo "Second argument does NOT equal 'Student' ( = )."

fi
```

```
manahil@ubuntu:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
Second argument equals 'Student' ( = ).
manahil@ubuntu:~$ ./setup.sh 10 Test
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
Second argument does NOT equal 'Student' ( = ).
manahil@ubuntu:~$ _
```

```
if [ "$str" != "Student" ]; then

          echo "Second argument is not equal to 'Student' ( != )."

  else

            echo "Second argument equals 'Student' ( != false)."

fi
```

```
manahil@ubuntu:~$ ./setup.sh 10 Test
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
Second argument does NOT equal 'Student' ( = ).
Second argument is not equal to 'Student' ( != ).
manahil@ubuntu:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
Second argument equals 'Student' ( = ).
Second argument equals 'Student' ( != false).
manahil@ubuntu:~$ _
```

```
if [ -z "$str" ]; then

          echo "Second argument is empty (zero-length)."

  else

            echo "Second argument is not empty."

fi
```

```
manahil@ubuntu:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
Second argument equals 'Student' ( = ).
Second argument equals 'Student' ( != false).
Second argument is not empty.
manahil@ubuntu:~$ ./setup.sh 10
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
Second argument does NOT equal 'Student' ( = ).
Second argument is not equal to 'Student' ( != ).
Second argument is empty (zero-length).
manahil@ubuntu:~$ _
```

---

6. Script: For Loop and Argument Handling

Scenario:

Write a script that prints all provided arguments and demonstrate correct handling of quoted multi-word arguments.

Steps:

1. Create/overwrite setup.sh to print every argument using "$@" in a for loop and save the file.

```
#!/bin/bash
#
## Print all arguments using "$@"
#
echo "Printing all arguments using \"\$@\":"
#
for arg in "$@"; do
  echo "Argument: $arg"
#
  done
```

2. Run the script with mixed single and quoted multi-word arguments and capture the output showing each argument on its own line.

```
"setup.sh" 11L, 153B written
manahil@ubuntu:~$ chmod +x setup.sh
manahil@ubuntu:~$ ./setup.sh one "two words" three
Printing all arguments using "$@":
Argument: one
Argument: two words
Argument: three
manahil@ubuntu:~$ _
```

---

7. Script: While Loop Summation and Functions

Scenario:

Implement an interactive or non-interactive summation function and a demonstrated function that returns a numeric result.

Steps:

1. Write an interactive while-loop that accumulates numbers until q is entered and shows running totals.

```
#!/bin/bash_
~
```

```
manahil@ubuntu:~$ chmod +x setup.sh

manahil@ubuntu:~$
manahil@ubuntu:~$ ./setup.sh
manahil@ubuntu:~$ _
```

2. Add a function that accepts two numeric arguments,

```
#!/bin/bash
sum=0

while true; do

          read -p "Enter a number (or 'q' to quit): " input

            if [ "$input" = "q" ]; then

                        break

                          fi

                            sum=$((sum + input))

                              echo "Total Score: $sum"

                      done

                      echo "Final total: $sum"_
~
```

```
manahil@ubuntu:~$ ./setup.sh
Enter a number (or 'q' to quit): 5
Total Score: 5
Enter a number (or 'q' to quit): 7
Total Score: 12
Enter a number (or 'q' to quit): 8
Total Score: 20
Enter a number (or 'q' to quit): q
Final total: 20
manahil@ubuntu:~$
```

3. returns their sum, and demonstrate capturing its result in a variable.

```
                      echo "Final total: $sum"
sum_args() {

          a=$1

            b=$2

              echo $((a + b))

      }


      # Demonstration
      #
      _echo "Now demonstrating sum_args function:"
      #
       result=$(sum_args 3 4)
      #
       echo "sum_args(3,4) returned: $result"
      #
```

```
"setup.sh" 41L, 517B written
manahil@ubuntu:~$ ./setup.sh
Enter a number (or 'q' to quit): q
Final total: 0
Now demonstrating sum_args function:
sum_args(3,4) returned: 7
manahil@ubuntu:~$ _
```
